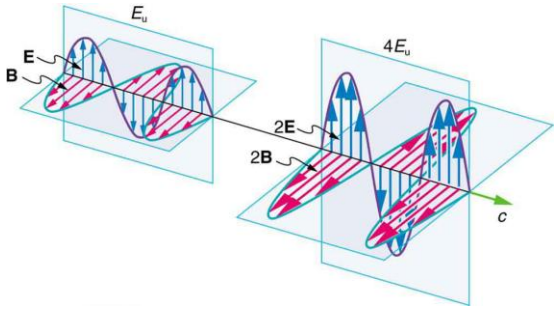
Chapter 55 – EM Wave Intensity
Calculus Physics II

Energy Carried by Electromagnetic Waves – Because EM waves are Electric and magnetic fields, they carry energy, this electric and magnetic energy is what is transported by the wave.

- Electromagnetic waves carry energy by way of their electric and magnetic fields.
- There is equal amount of energy in the magnetic and electric fields.
- The higher the frequency of the EM wave the more energy the wave has stored its electric and magnetic field.

$$Energy\ in\ EM\ waves = hf$$

h = 6.63 x 10^{-34} Js is called Planck's constant
f – frequency of wave in Hz
c – speed of light in vacuum c = 3 x 10^{8} m/s



- These oscillating fields can exert forces on charges in a system and do work on them, transferring energy.

Relationship between $\vec{E}$ & $\vec{B}$ – The strength of the electric and magnetic fields in EM waves are related by the speed of light in vacuum (c). If one field strength is known, then the other can be determined by

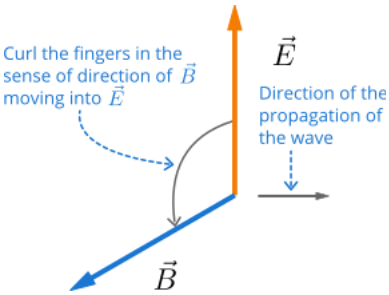
$$|\vec{E}|^2 = \frac{|\vec{B}|^2}{\varepsilon_o \mu_o} \qquad |\vec{E}| = c|\vec{B}| \qquad c = \frac{|\vec{E}|}{|\vec{B}|} = \frac{1}{\sqrt{\varepsilon_o \mu_o}}$$

$\vec{E}$ – Electric field strength $\vec{B}$ – Magnetic field strength
c – speed of light = 3 x 10^{8} m/s μ_o = 1.257 x 10^{-6} Tm/A ε_o = 8.85 x 10^{-12} C^2/Nm^2

The Poynting Vector ($\vec{S}$) – The power per area and direction of a EM wave is given by the Poynting Vector. The Poynting vector is perpendicular to both the Electric and Magnetic field and points in the direction of the propagation of the wave. The magnitude of the Poynting vector is the power transported through an area for a EM wave. Units are Watts/m^2

$$\vec{S} = \frac{1}{\mu_o}\vec{E}x\vec{B} \qquad |\vec{S}| = \frac{1}{\mu_o}|\vec{E}||\vec{B}|sin(\varphi)\ \ since\ \varphi = 90°$$

$$|\vec{S}| = \frac{1}{\mu_o}|\vec{E}||\vec{B}|$$



- The direction of travel of an EM wave is given by the cross product of the Electric Field with the Magnetic field. (Right hand rule!)

Average Intensity (I_{avg} or I) – The total amount of energy per unit time in the EM fields that pass through an area (Average magnitude of the Poynting vector). Intensity is the total energy per unit time (Power) per unit area (A) carried by an electromagnetic wave or the Power (P) per unit area. Units are W/m^2

$$I_{avg} = \left|\vec{S}\right|_{avg} = \frac{(\text{Total Energy/time})}{\text{Area}} = \frac{\text{Power}}{\text{Area}}$$

$$\left|\vec{S}\right| = \frac{1}{\mu_o}\left|\vec{E}\right|\left|\vec{B}\right| \qquad \text{and} \qquad \left|\vec{E}\right| = c\left|\vec{B}\right|$$

Note: the magnitudes of the electric and magnetic fields used are the rms values, to get the peak value of the E or B field

$$E_o = \sqrt{2}\, E_{rms} \qquad B_o = \sqrt{2}\, B_{rms}$$

The energy in the EM wave is in the electric and magnetic fields. The Energy in the magnetic field is equal to the energy in the electric field.

$I_{avg} = (c\varepsilon_o E^2) = (cB^2/\mu_o)$, where E and B are rms values of the field strength

$$I_{avg} = c\varepsilon_o E^2 = \frac{E^2}{c\mu_o} \qquad I_{avg} = \frac{cB^2}{\mu_o}$$

Example: For light incident on a movie screen the magnetic field strength is 1.33×10^{-6} T and the electric field strength is 400 V/m what is the intensity of the light?

$I = c\varepsilon_o E^2 = 420\ W/m^2$ $\qquad$ $I = cB^2/\mu_o = 420\ W/m^2$

Example: The intensity of light on a 2 m^2 solar panel is 1,000 W/m^2, what are the values of the peak magnetic and electric fields and what are their rms values?

$I = c\varepsilon_o E^2$ therefore the rms values of the field are

$E = (I/c\varepsilon_o)^{1/2} = 614$ V/m and $B = E/c = 2.05 \times 10^{-6}$ T

And the peak values are

$E_o = \sqrt{2}\, E_{rms} = 868\ V/m$ $\qquad$ $B_o = \sqrt{2}\, B_{rms} = 2.9\, x\, 10^{-6}\, T$

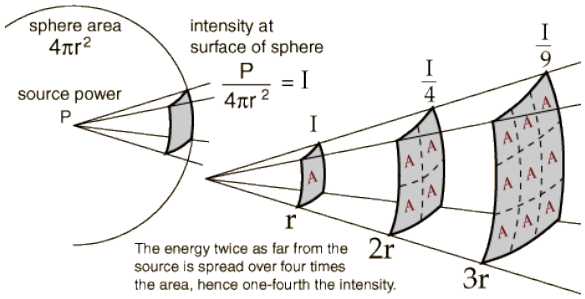
Example: If an EM wave has an electric field with a magnitude of 400 V/m what is the magnitude of the magnetic field?

$$\left|\vec{E}\right| = c\left|\vec{B}\right| \qquad B = E/c = 1.33 \times 10^{-6}\ T$$

Intensity of a point source – For an isotropic point source of EM waves (like a light bulb) the Intensity is given by the amount of power emitting from the source divided by the area around the source, this area is a spherical surface and is equal to $A = 4\pi r^2$. Therefore, the Intensity of a point source is given by

$$I = \frac{P}{A} = \frac{P}{4\pi r^2}$$

- The intensity of light falls off as $1/r^2$ therefore the intensity at twice the distance (2r) would have a quarter of the intensity and at 3r would have 1/9 the intensity.



Example: For an isotropic 100 W light bulb, determine the intensity at 1 m from the bulb?

$$I_{1m} = \frac{P}{4\pi r^2} = 8\ W/m^2$$

At 2 m what is I?

$$I_{2m} = \frac{P}{4\pi r^2} = 2\,W/m^2$$

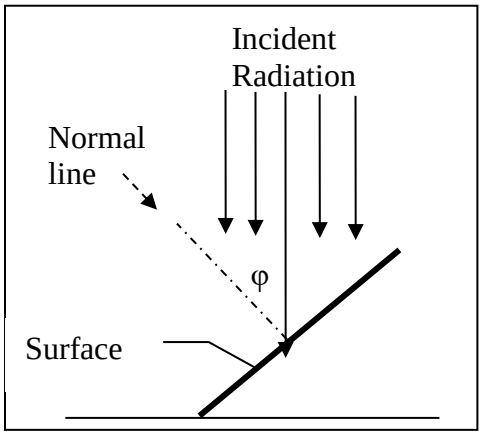
Example: If the power output of the sun is 3.9×10^{26} W and the earth is 1.5×10^{11} m from the sun, determine the intensity of the light from the sun at the earth?

$$I = P/(4\pi r^2) = (3.9 \times 10^{26}\ W)/(4\pi(1.5 \times 10^{11}\ m)^2) = 1{,}400\ W/m^2$$

Power (P) incident on surface – if the normal to the surface is in the same as the incident rays then the maximum intensity is incident on the surface but if the normal to the surface is at an angle to the incident rays then less of the area is exposed to the rays by a factor of cosφ.

$$P = IA\cos\varphi$$

P is the Power,
I is the intensity,
A is the area and
Angle φ is measured from the incident radiation and the normal to the surface.



Efficiency (η) – The ratio of the useful energy/work performed by a device relative to the total energy/work supplied to the device.

$$\eta = \frac{\text{Useful output}}{\text{Total input}}\ x\ 100\%$$

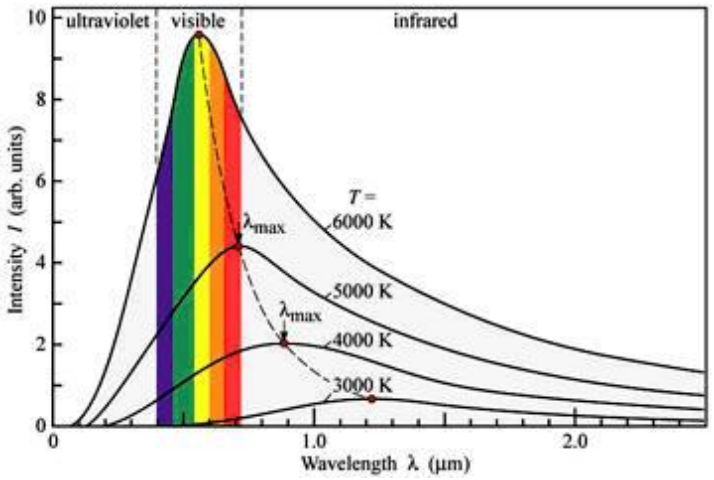
- Converting EM waves power into electrical power requires a solar cell. The efficiency of the solar cell indicates the amount of useful electrical power the cell outputs relative to the total EM wave power incident on the cell.

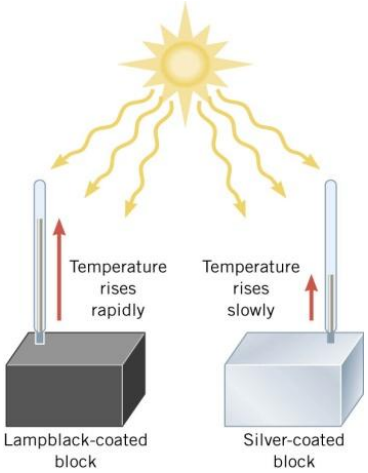
$$P_{output} = \eta[P_{Total}] = \eta[IA\cos\varphi]$$

Blackbody radiation - is defined as a prefect emitter and absorber of all electromagnetic waves. (in general, a blackbody is a solid material, not gas)

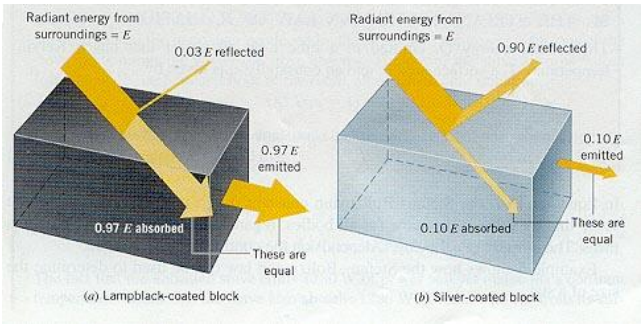
- ✓ A good emitter is a good absorber.
- ✓ A poor emitter is a poor absorber.

- Any object that has an internal energy (temperature) emits a spectrum of EM radiation.
- The hotter the object the more EM energy it emits and the shorter wavelength (higher frequency) EM radiation it emits.
- The hotter the object the more EM radiation becomes a significant energy loss.
- When more radiation is absorbed by an object than emitted by the object then the body gets hotter.
- When more radiation is emitted by an object than absorbed by the object then the body gets cooler.





Emissivity (e) is the ratio of the energy an object actually radiates to the energy the object would radiate if it were a perfect emitter.



Emissivity is a number between 0 and 1,

- e = 0 there is no EM radiation emitted or absorbed by the object
- e = 1 being a perfect emitter and absorber. For a particular region of the EM spectrum.

The peak wavelength (λ_{peak}) (in meters) of the Blackbody radiation only depends on the temperature (T) of the body (in degrees Kelvin) and is given by

$$\lambda_{peak} = \frac{2.898 \times 10^{-3}\ \text{mK}}{T}$$

Example: If the peak wavelength of the sun is 550 nm (green) what is the temperature on the surface of the sun in kelvin?

$$\lambda_{peak} = \frac{2.898 \times 10^{-3}\ \text{mK}}{T} \text{ therefore } T = \frac{2.898 \times 10^{-3}\ \text{mK}}{\lambda_{peak}} = 5{,}270\ \text{K}$$

Black body radiation for different temperatures

Temperature	Peak EM wavelength emitted (EM waves)	
250 K (- 23C)	12 microns	Ice - freezer
300 K (27 C)	10 microns	Human body
500 K (227 C)	6 microns	Temp. of oven
800 K (527 C)	3 microns	faint glow element
1500 K (1,227 C)	2 microns	red glow
3000 K (2,727 C)	1 microns	yellow glow
4500 K (4127 C)	0.5 microns	white light

Color temperature – the measure of the "whiteness" of the light using the color temperature (measured in Kelvin) of the emitted light. Color temperature is based on blackbody radiation.

2500 K yellowed light
3400 K whiter light
5000 K bright white light



The Stefan-Boltzmann Law of Radiation - The radiant energy Q, emitted in a time Δt, by an object that has a Kevin temperature T, a surface area A and an emissivity e is given by

$$Q = e\sigma T^4 A\Delta t$$

σ is the Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8}$ J/(sm^{2}K^4)
e is the Emissivity.
A is the area of what is radiating.
T is the temperature in Kelvin! $T = T_K = 273 + T_c$

Where T_K is the temperature in Kelvin and T_c is the temperature in degrees Celsius.

Radiant Power – the power or amount of energy per second emitted from a blackbody.

$$P = Q/\Delta t = e\sigma A T^4$$

- Temperature is always measured from absolute zero and in Kelvin.

Net radiant power is the power the object emits minus the power it absorbs. When an object has a temperature higher than its surroundings the object emits a net radiant power P_{net}.

$$P_{net} = Q/\Delta t = e\sigma A(T^4 - T_o^4)$$



T = temperature of object in Kelvin
T_o = temperature of environment in Kelvin

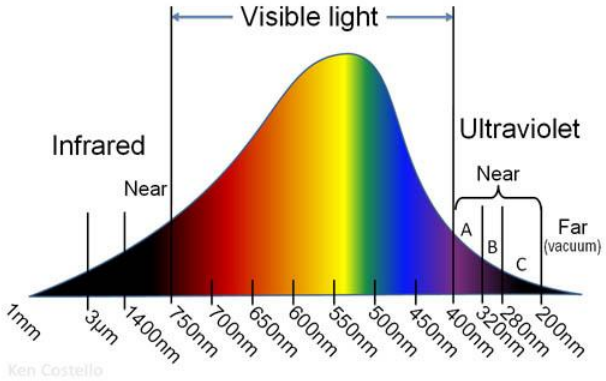
- Object in thermal equilibrium if same temperature as environment and there is no net radiation emitted.
- Objects that are warmer than their environment emit infrared radiation (lose energy)
- Objects that are cooler than their environment absorb infrared radiation (gain energy)

Example: Bob turns on his stove element that has an area of 0.02 m^2 and notices that the heating element is glowing red and has a temperature of 1,000° C (1,273 K), how much power is being emitted in radiation if the room temperature is 25° C (298 K)? Assume the Emissivity of the element is e = 0.9

$P_{net} = Q/\Delta t = e\sigma A(T^4 - T_o^4) = 2{,}672$ W

Solar Radiation – The electromagnetic radiation of the sun is peaked in the visible part of the EM spectrum. The peak is green (wavelength of 550 nm) which corresponds to the maximum sensitivity of the human eye.

- There is a significant portion of solar radiation in the infrared.
- The infrared is felt as the warmth from the sun and heats the surface of the earth.
- The sun's radiation is very close to a black body emitter at a temperature of 5,800 K.



The Sun and its Intensity – The total intensity emitted by our Sun across the EM spectrum is about 6.33×10^7 W/m^2. This energy is due to the temperature of the sun which has a surface temperature of around 5,780 K.

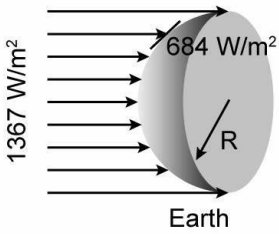
- The energy output of the sun is very stable, it only changes by 0.25% over time and the change is mostly due to sunspot formation.

  Sunspots are temporary areas on the sun that are at lower temperature than the surface, the lower temperature is caused by very strong magnetic fields restricting the flow of the material.

The power emitted in the form of EM waves from the Sun is $P_{Sun} = 3.865 \times 10^{26}$ W, the Sun's rays can be considered parallel when they reach the earth and assuming the distance from the sun to earth is 1.5×10^{11} m, the amount of solar Intensity reaching the earth is approximately.

$$I = \frac{P}{4\pi r^2} = 1400\ W/m^2$$

The actual intensity of the Sun's EM waves reaching the earth is give as the Solar constant (I_{sc}) and is equal to

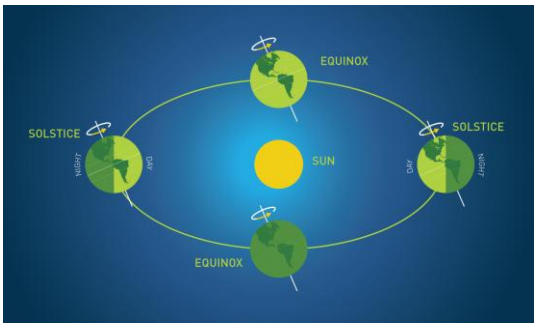


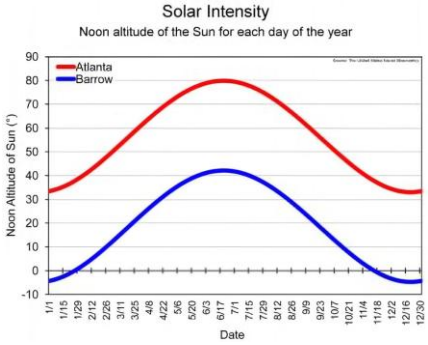
Solar Constant I_{sc} = 1,367 W/m^2

- ❖ This is the Solar Intensity before it is incident on the earth's atmosphere and does not account for loss due to atmosphere or angle to the earth's surface.

A rough estimate of loss of solar intensity due to the earth's atmosphere and scattering form clouds, particulates and reflection off the earth's surface is about 30%. An average value for the amount of the sun's intensity at the surface of the earth on a clear day is about **1,000 W/m^2.**

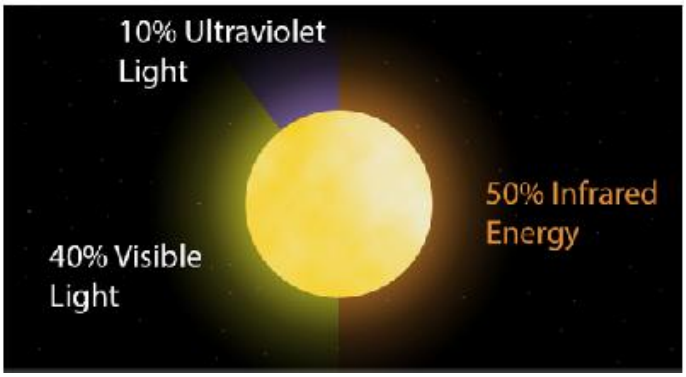
The sun changes angle relative to the earth's surface throughout the day, this changes the solar intensity at the surface of the earth. The maximum intensity is when the sun is directly overhead. The Sun's Intensity changes due to the relative change of the earth's surface throughout the year and depending on the latitude of the location, this changes the solar Intensity and is shown in the graph below. The maximum intensity is the summer solstice, the longest day of the year that is in June and minimum in December (Winter solstice).





EM Spectrum from the Sun's surface

- 43 % is in the Visible range
- 49 % is in the Infrared range
- 7 % is in the ultraviolet range
- Less than 1% is in the gamma ray, x-ray and radio wave ranges



The graph shows 1) the outline of an ideal blackbody emitter at 5900 k along, 2) Extraterrestrial solar radiation - the EM spectrum that reaches the earth from the sun, 3) Terrestrial solar radiation – the EM spectrum that reaches the surface of the earth.

EM radiation range of sun

- Before reaching earth the EM spectrum is approximately 200 nm (UVA) to 4000 nm (IR).
- At surface of earth the EM spectrum is approximately 300 (UVB) to 1800 nm (IR) with absorption in other regions.

